

UMT503: AUTOMATIC CONTROL SYSTEMS

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Course Contents

Basic Concepts

Open and closed loop control systems, dynamics of mechanical systems and electromechanical systems (motors, gears), electric circuit models, heat and fluid-flow models, concept of transfer function, block diagrams and signal flow graphs

Control hardware and their models

Potentiometers, synchros, LVDT, dc and ac servomotors, tacho-generators, and stepper motors.

Analysis

Transient analysis of second order systems, Time domain specifications, Steady-state errors and error constants, Concepts of P, PD, PI and PID types of control and their realizations

Stability

Definition, Routh-Hurwitz criterion, Root locus techniques, Nyquist criterion, Bode plots, Relative stability, concepts of gain and phase margins.

Feedback Analysis

The basic equations of control-Stability, Tracking, Regulation and Sensitivity, Control of steady-state errors for polynomial inputs, system type for tracking, regulation and disturbance rejection.

Compensation

Lead, Lag and lag-lead compensators, Design of compensating networks for specified control system performance.

State Space Analysis

Concepts of state, State variables and state models, State space equations, transfer function, Transfer model, State space representation of dynamic systems, State transition matrix, Decomposition of transfer function, Controllability and observability

RECOMMENDED BOOKS

- Control Systems Engineering by **I.J. Nagrath and M. Gopal**, New Age
- Automatic Control Systems by **B.C. Kuo and F. Golnaraghi**, Wiley
- Linear Control Systems with MATLAB Applications by **B.S. Manke**, Khanna Publisher
- Modern Control Engineering by **K. Ogata**, PHI

ASSESSMENT SCHEME

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (Assignments//Quizzes/Lab Evaluations)	35